Irtifa Haider

irtifa.stu2019@juniv.edu irtifahaider Irtifa Haider Ahona IrtifaHaider Irtifa Haider

RESEARCH INTEREST

Information Retrieval, Text and Graph Mining, Social Network Analysis, Clustering and Machine Learning, Large Language Models (LLMs) and Trustworthy NLP

EDUCATION

Master of Science (M.Sc.) in Computer Science and Engineering **Sept 2025 – Present**
Jahangirnagar University, Savar, Dhaka, Bangladesh

- **Relevant coursework:** Natural Language Processing, Machine Intelligence and Expert Systems, Advanced Neural Network, Distributed Database, Network Security.

Bachelor of Science (B.Sc.) in Computer Science and Engineering **Mar 2020 – Sept 2025**
Jahangirnagar University, Savar, Dhaka, Bangladesh

- **CGPA:** 3.42/4.00
- Thesis: *The Star Paradigm: Redefining Social Network Clustering*.

RESEARCH EXPERIENCE & PUBLICATIONS

- **Haider, I.**, Alam, S., Hossan, M. T., Anwar, M. M., & Aurpa, T. T. **A Comprehensive Text Optimization Approach to Bangla Summarization.** *Proceedings of the Second Bangla Language Processing Workshop (BLP), co-located with the 14th International Joint Conference on Natural Language Processing (IJCNLP–AACL)*, 2025. *Presented at the conference.*
- **Star-Based Community Detection in Social Networks: The Hub Effect and a Share-Rule Remedy.** *Manuscript in preparation.*
  - Applied the Star Clustering Algorithm to community detection: high-degree nodes act as cluster centres, and the number of communities is not set in advance.
  - On Zachary's Karate Club, achieved the highest F1-score (0.85) and modularity (0.29) against K-Means, K-Medoids, and Agglomerative Clustering.
  - Studies an incremental setting with streaming node arrivals, along with sensitivity to parameter τ, node-arrival order, scalability, and robustness to noisy edges.
- **The Patient Insists: Measuring Sycophancy and Safety Drift of LLMs in Multi-turn Bangla Medical Dialogue.** *Manuscript in preparation.*
  - Measuring how large language models yield to repeated patient pressure and drift from safe medical advice over multi-turn conversations in Bangla, a low-resource language.
- **HSECRS: Hybrid Smart Energy Control and Redistribution System.** *Manuscript in preparation.*
  - Developing an AI-enabled smart energy management and redistribution system, with an IoT prototype that optimizes power distribution across solar, grid, and battery sources while prioritizing critical loads.

PROFESSIONAL EXPERIENCE

QA Functional Tester (Internship) | A1QA **April 2026 – June 2026**

- Worked with various testing types including Smoke Testing, Acceptance Testing, and Manual Testing.
- Designed and executed test cases, performed test planning, identified bugs, and reported defects using Jira with proper documentation.

Coding Instructor | CoderPals **Sept 2024 – April 2026**

- Taught Scratch and Python through game-based and project-based lessons for children.
- Simplified programming concepts to foster computational thinking, creativity, and problem-solving skills.

ACADEMIC PROJECTS

Plant Classification | Python, scikit-learn, scikit-image, NumPy, pandas, matplotlib, joblib

- Built a machine-learning pipeline to classify 18 plant types using HOG features and SGD/SVM classifiers.
- Achieved 80% classification accuracy through feature extraction and model tuning.

LivSmart: Smart Living Community App | Java, XML, Firebase, Dokka, JUnit, Mockito, MVVM

- Developed an Android application for residential services (service requests, payments, events, alerts) with a Firebase backend for dashboards, parking, and amenities booking.

OpenDoor: Property Rental & Purchase App | Java, XML, Firebase, CircleCI, JUnit, Mockito, MVVM

- Built an Android app for property rental and purchase with aggregated listings, filtering, real-time alerts, and a mortgage calculator.

IT Training Hub | HTML, CSS, JavaScript, PHP, MySQL, MVC

- Developed a web platform for managing IT training courses, trainers, and participants.
- Applied the Model–View–Controller (MVC) architecture to organize the application and support maintainability.

Early Programming Projects | C, C++

- **Learn and Teach Programming**: A learning platform with lessons to support programming practice.
- **Criss Cross Game & Calculator:** A two-player Tic-Tac-Toe with a calculator for arithmetic operations.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript

AI Frameworks & Models: TensorFlow, PyTorch, scikit-learn, NumPy, pandas, matplotlib, Word2Vec, BERT, BanglaBERT, mBERT, XLM-RoBERTa (XLM-R), mT5, T5, GPT-4, Gemini, Llama, Qwen2.5

Web Development: HTML, CSS, React, Bootstrap, Django

Development Tools: Git, GitHub, Android Studio, VS Code, NetBeans, Eclipse, Anaconda, Visual Paradigm, CI/CD, JUnit, Mockito, MATLAB, Slack, Trello, MySQL, Firebase, Blynk

Productivity & Collaboration Tools: Microsoft Office Suite (Word, Excel, PowerPoint), Toggl, Overleaf, JIRA

CERTIFICATIONS

- **Certificate of Attendance** – IJCNLP–AACL 2025 **Jan 2026**
- **Cython Codecraft: Data Structure & Algorithm Bootcamp** – bongoDev, 3-Day Bootcamp **Feb 2024**
- **Learn & Hack Web Development Bootcamp** – bongoDev, 2-Week Bootcamp **Jan 2024**
- **Legacy JavaScript Algorithms and Data Structures** – freeCodeCamp **Aug 2023**
- **Responsive Web Design** – freeCodeCamp **Jul 2023**

HONORS & AWARDS

- **International Collegiate Programming Contest (ICPC):** Competed in the preliminary round. **2021**
- **National Collegiate Programming Contest (NCPC):** Participated in the preliminary round. **2021**
- **Competitive Programming:** Solved 400+ problems on Codeforces (*Irtifa_CSE-29*), AtCoder (*Irtifa_CSE29*), and LeetCode (*haiderIr*).
- **Merit Scholarships:** Government Board Merit Scholarships in both *JSC (2014)* and *SSC (2017)*.

VOLUNTEER EXPERIENCE

Volunteer | Deep Ecology and Snake Conservation Foundation **2023 – Present**

- Participate in environmental conservation initiatives, awareness campaigns, and community outreach programs.

Event Support Staff | National Collegiate Programming Contest (NCPC), Jahangirnagar University **2024**

- Assisted in coordinating on-site logistics and the smooth execution of NCPC 2024.
- Supported venue management and communication with committees to ensure successful event operations.